

Part B

Across its 44 acute and day hospitals, STADIOcare Group is having trouble accurately forecasting and controlling patient demand and hospital capacity. Over the previous two years, average bed occupancy has decreased from 71.5% to 69.2% and has become more erratic. There is pressure on available beds and resources when periods of low occupancy are followed by abrupt spikes in demand.

Increasing and fluctuating patient length of stay complicates the issue even more. For comparable procedures, the average length of stay has increased from 3.9 to 4.2 days, with considerable variation across hospitals. This makes it challenging to efficiently plan staffing and forecast bed availability. As a result, agency spending and nurse overtime have risen from R610 million to R890 million.

Additionally, STADIOcare is combining data from HR, billing, appointments, admissions, and other systems into a group-wide data platform. Nevertheless, the organization's capacity to effectively use its data for proactive decision-making is constrained by the inconsistent data quality and historical coverage.

The main issue is that STADIOcare currently lacks an adequately data-driven method to forecast patient demand, bed utilization, and length of stay in order to facilitate proactive workforce and capacity planning. This leads to increased labor costs, operational strain, and inefficient use of resources. By analyzing past patient, operational, and workforce data, a data science project can solve this issue by predicting bed utilization, identifying demand patterns, and identifying factors linked to longer stays. The ensuing insights could help STADIOcare achieve its strategic goal of increasing sustainable occupancy and becoming a more data-driven organization while supporting more proactive capacity and staffing decisions.